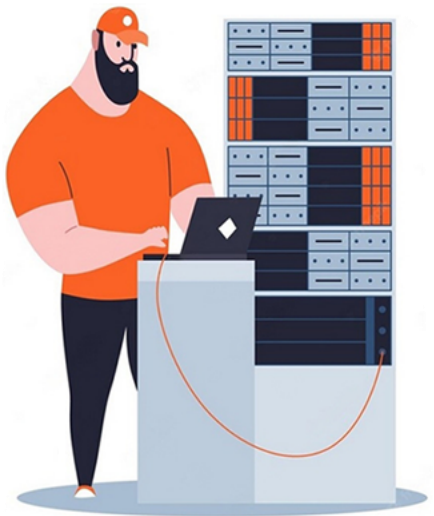
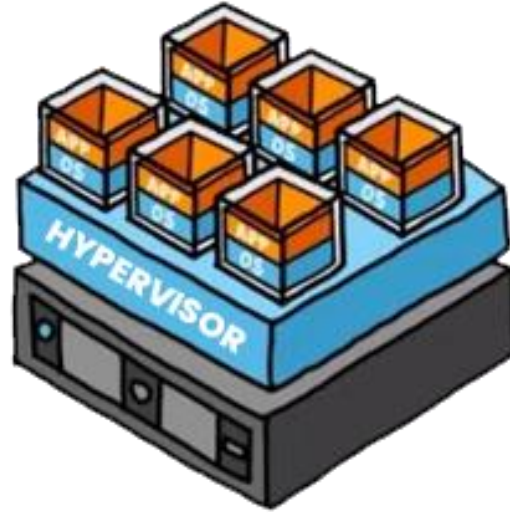


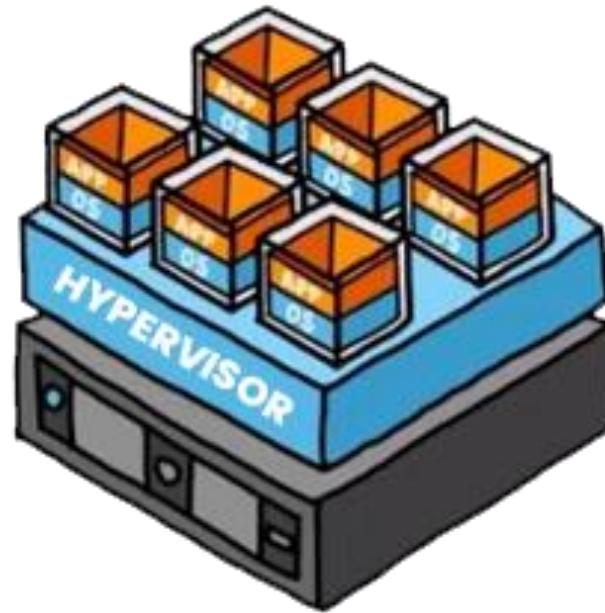
AGENDA

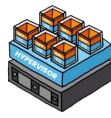
About Virtualization.
Type of Virtualization.
Vendor in Virtualization Industry.
About VMware.
VMware Certification.
Benefits of Certification.
Certification Requirement.
About VMware Products.



INTRODUCTION
OF COURSE

VIRTUALIZATION





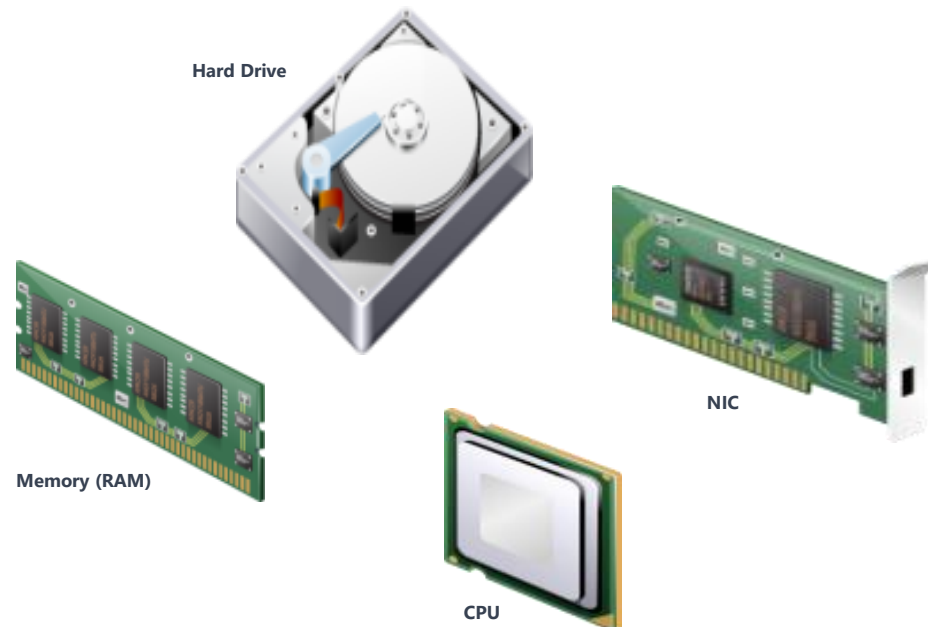
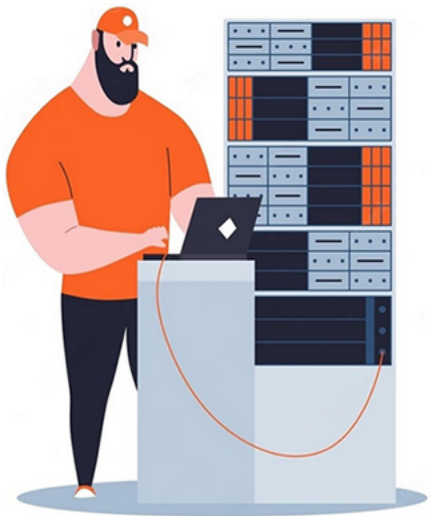
Virtualization refers to the creation of a virtual (rather than actual) version of something, such as an operating system, a server, a storage device, or network resources.

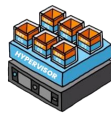
It is a technology that allows multiple virtual instances of resources to be created on a single physical hardware system, effectively enabling the sharing of physical resources between multiple users or applications.

Intel developed firstly V.T. (Virtualization Technology) in Dual Core processor family.

Virtualization has become fundamental in IT infrastructure for improving efficiency, scalability, flexibility, and resource utilization. It is commonly used in data centers, cloud computing, development environments, and testing.

Virtualization is totally cost-effective computer-based technology.





Hardware Virtualization (Server Virtualization)

Hardware virtualization allows multiple virtual machines (VMs) to run on a single physical host machine, each with its own operating system (OS).

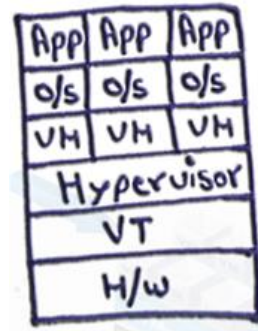
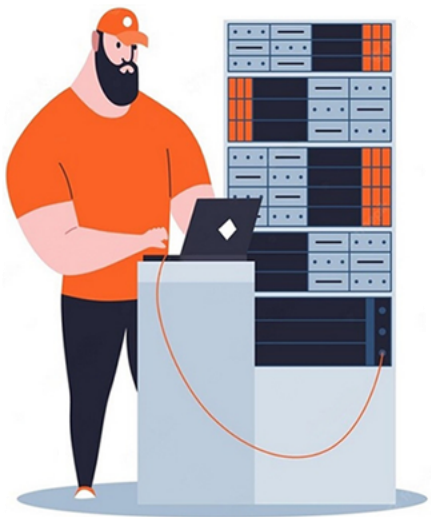
How it works:

A hypervisor (also called a Virtual Machine Monitor or VMM) is used to create and manage virtual machines. The hypervisor sits between the hardware and the operating system, isolating VMs and allowing them to operate independently.

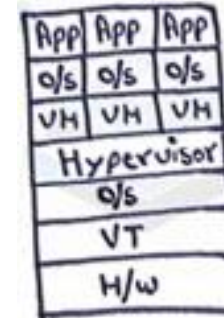
Types of Hypervisors:

Type 1 Hypervisor (Bare-Metal): Runs directly on the physical hardware (e.g., VMware ESXi, Microsoft Hyper-V, Xen).

Type 2 Hypervisor (Hosted): Runs on top of an existing operating system (e.g., VMware Workstation, Oracle VirtualBox).



Bare-Metal



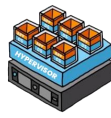
PARA-Hosted

Benefits:

- Better hardware utilization.
- Isolation of applications and environments.
- Easier backup, migration, and scaling.
- Running multiple OSes on the same physical server.

INTRODUCTION
OF COURSE

OPERATING SYSTEM VIRTUALIZATION

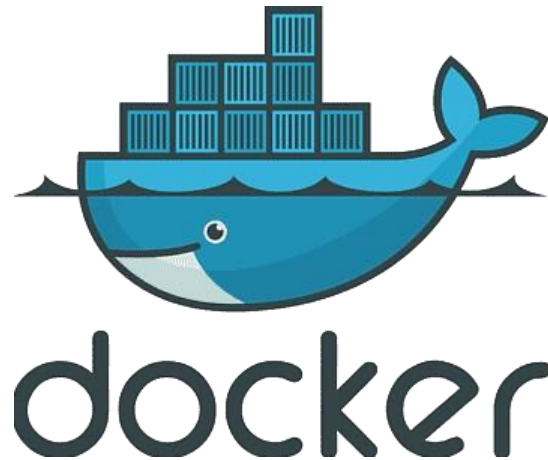


Operating System Virtualization (Containerization)

Operating system-level virtualization, often referred to as containers, allows multiple isolated user-space instances to run on a single OS kernel. Unlike VMs, containers share the same underlying OS kernel but remain isolated from each other.

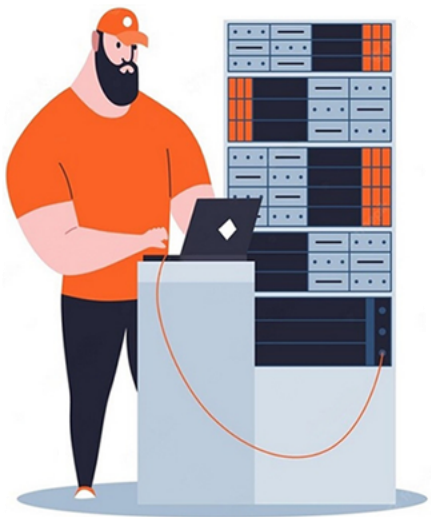
How it works:

A container runtime (e.g., Docker, Podman) is used to create, deploy, and manage containers. Each container runs a specific application or service along with its dependencies, making it portable and lightweight.



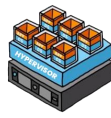
Benefits:

- Faster startup times compared to VMs.
- Efficient resource usage due to shared kernel.
- Simplified deployment and management of applications in microservices architecture.



INTRODUCTION
OF COURSE

NETWORK VIRTUALIZATION

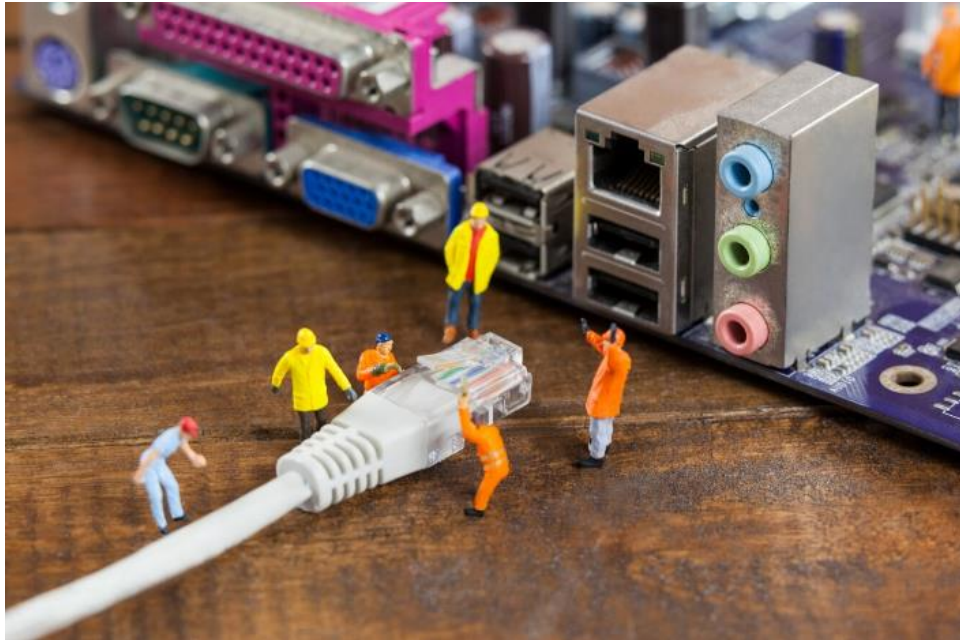
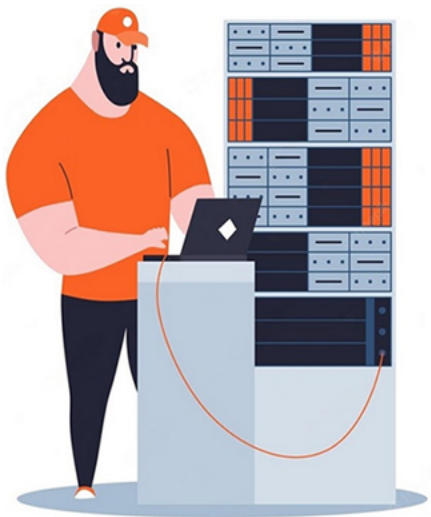


Network Virtualization

Network virtualization abstracts physical network resources to create virtual networks. This allows network resources like bandwidth, IP addresses, and routing to be allocated dynamically based on demand.

How it works:

Network virtualization enables the creation of virtual networks that run on top of physical networks. Technologies like Software-Defined Networking (SDN) and Network Functions Virtualization (NFV) allow the decoupling of network control and data planes.

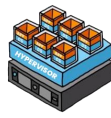


Benefits:

- Improved network flexibility and scalability.
- Easier network management.
- Cost-effective, especially in cloud environments.

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STORAGE VIRTUALIZATION

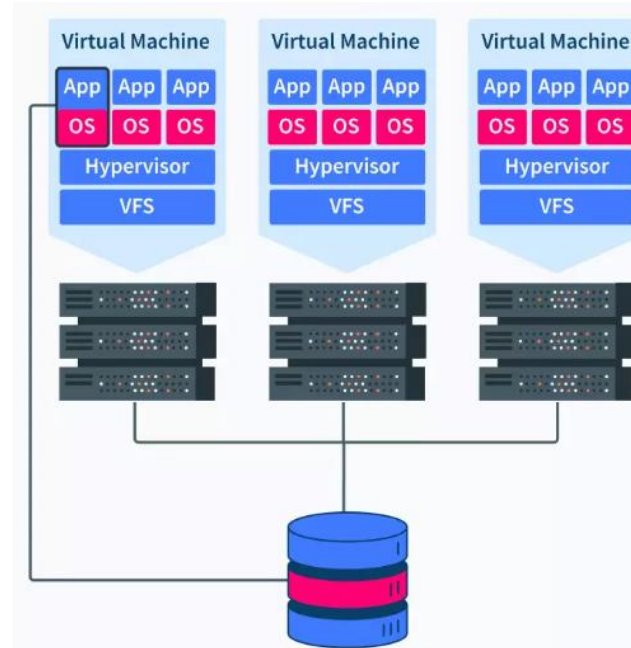


Storage Virtualization

Storage virtualization involves abstracting the physical storage resources into a virtual pool, which can then be managed as a single entity.

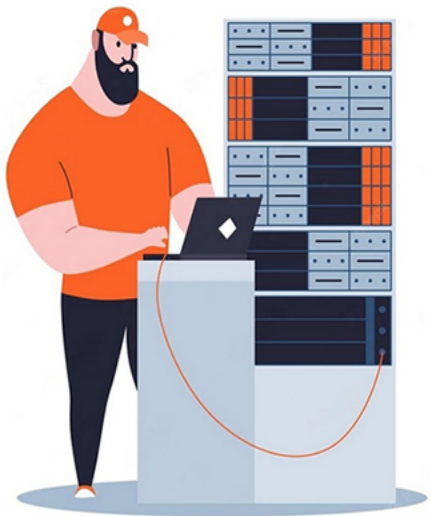
How it works:

Storage virtualization combines multiple physical storage devices into one virtual storage system, which can be managed and accessed as a single unit.



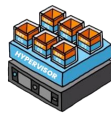
Benefits:

- Simplified storage management.
- Easier to scale and allocate storage dynamically.
- Increased flexibility in provisioning and managing storage resources.



INTRODUCTION
OF COURSE

DESKTOP VIRTUALIZATION

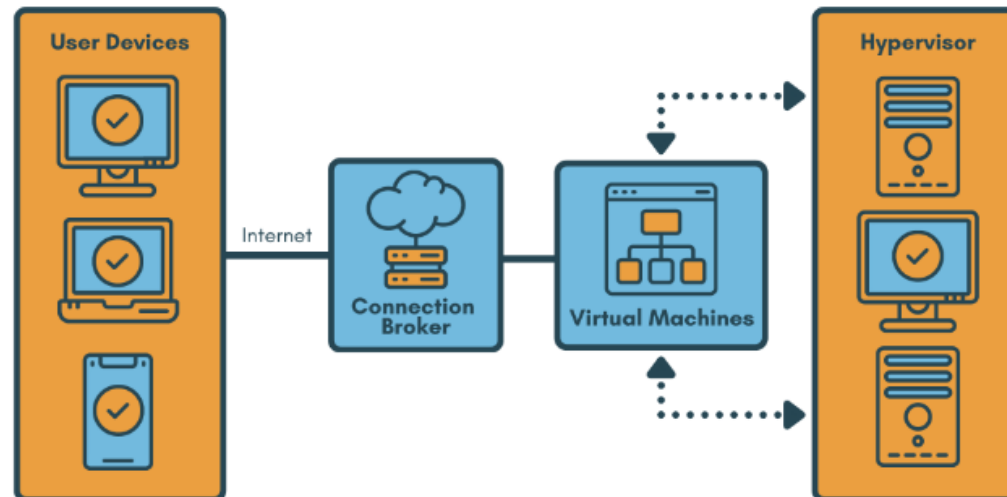
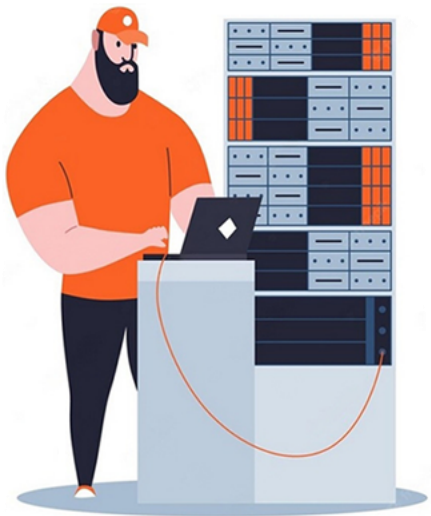


Desktop Virtualization

Desktop virtualization allows users to access their desktop environments remotely via a virtualized instance.

How it works:

Virtual Desktop Infrastructure (VDI) or Desktop-as-a-Service (DaaS) solutions allow organizations to host desktop environments on virtual machines or servers in data centers or the cloud.

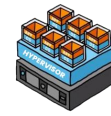


Benefits:

- Centralized management of desktops.
- Remote access for users.
- Improved security and data control.

INTRODUCTION
OF COURSE

VENDOR's IN VIRTUALIZATION



Server Virtualization Vendor's



VMware

- Products: VMware vSphere, VMware Workstation, **VMware ESXi**
- VMware is one of the most well-known and widely adopted providers of virtualization solutions. vSphere is their enterprise-level server virtualization platform, and ESXi is the hypervisor used for running virtual machines.



Microsoft

- Products: **Microsoft Hyper-V**, Windows Server
- Hyper-V is Microsoft's virtualization technology, available in both server and desktop versions. It is integrated into the Windows Server product line and is also available as a standalone hypervisor.



Citrix

- Products: Citrix Hypervisor (formerly XenServer)
- Citrix's virtualization platform is based on the open-source **Xen hypervisor**. It is used for server and desktop virtualization, and Citrix also offers XenApp and XenDesktop for application and desktop virtualization.



Red Hat

- Products: **Red Hat Virtualization (RHV)**
- Based on the open-source KVM (Kernel-based Virtual Machine) hypervisor, Red Hat Virtualization is an enterprise solution designed for Linux and Windows environments.



Oracle

- Products: Oracle VM VirtualBox, Oracle VM Server for x86
- Oracle offers VirtualBox for desktop and development environments, and Oracle VM Server for x86 is used for server virtualization, primarily in enterprise environments using Oracle applications.



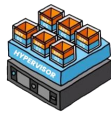
Nutanix

- Products: **Nutanix Acropolis Hypervisor (AHV)**
- Nutanix is known for its hyper-converged infrastructure, and Acropolis is the native hypervisor in Nutanix environments. It's designed to integrate with the company's storage and computing solutions.

INTRODUCTION
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ABOUT
VMware





About VMware

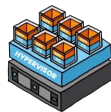
VMware is a global leader in virtualization and cloud computing technologies. Founded in 1998, VMware is based in Palo Alto, California. In 2004, VMware revolutionized the IT industry by pioneering virtualization software that allows multiple virtual machines (VMs) to run on a single physical server. Over the years, VMware has expanded its portfolio to cover a wide range of technologies, including cloud computing, network virtualization, storage, and more.



Industry	:	Computer Software
Founder	:	Diane Greene Mendel Rosenblum Scott Devine Ellen Wang Edouard Bugnion
Founded	:	Palo Alto, California, US, 1998, 24 years ago

INTRODUCTION
OF COURSE

CERTIFICATION

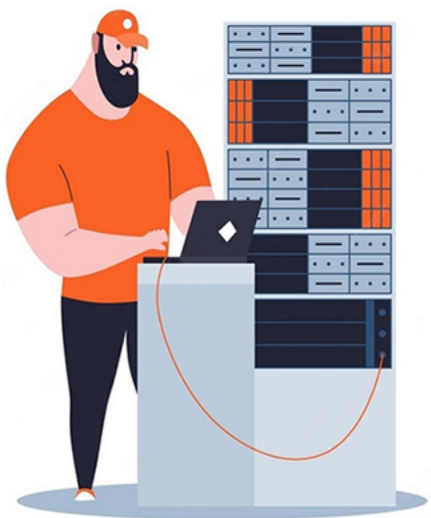


Benefits of Certification's

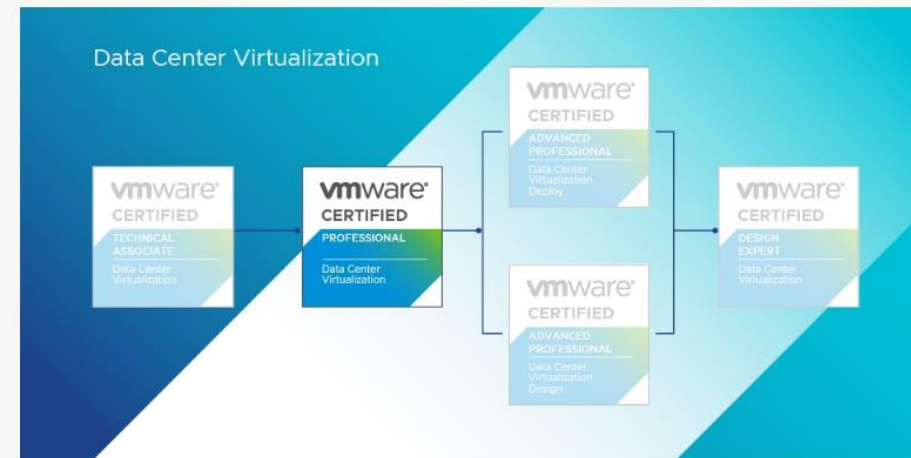
- Recognition of your technical knowledge and skills.
- Greater opportunities for career advancement.
- Official transcripts.
- Use of the certification logo in resume.
- Access to the exclusive portal & logo merchandise store.
- Invitation to beta exams and classes.
- Discounts from VMware Press.
- Discounted admission to VMware events.

VMware Certification

<https://www.broadcom.com/support/education/vmware/certification>



Data Center Virtualization Certification Path



Certification Path if you hold no VCP certifications

Expand All | Collapse All

STEP 1 : (Recommended) Gain experience with vSphere 7.x or vSphere 8.x



STEP 2 : (Recommended) Attend one of the training courses



STEP 3 : (Required) Pass the qualifying exam



VMware vSphere 8.x Professional (2V0-21.23)

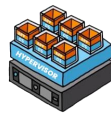
135 Minutes 70 Questions \$250 USD

[Schedule Exam](#)

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VMware
PRODUCT's



vSphere Suite

- ESXi Hypervisor
- vCSA

